Homework #2

7 points

Complete the assigned problems

Chapter 2: 1, 2, 5, 6, 7, 13-16, 18-22, 29, 31, 38, 39c, 42, 44-47, 51-53, 57 adh, 63 ace (scanned problems are at the end of this document)

Hint: See Chapter 2, Problem 63 Example Problem: **https://drive.google.com/file/d/1PKPWWMLUI_FUqZZr5Q16tQLhKYCMKWMG/view?usp=sharing**

Problem Solutions

Chapter 2: 1, 2, 5, 6, 7, 13-16, 18-22, 29, 31, 38, 39c, 42, 44-47, 51-53, 57 adh, 63 ace

1. AVR is a(n) _____-bit microcontroller.
2. The general purpose registers are _____ bits wide.
3. The value in LDI is _____ bits wide.
4. The largest number that can be loaded into the GPRs is _____ in hex.
5. What is the result of the following code and where is it kept?

```
LDI     R20,$15
LDI     R21,$13
ADD     R20,R21
```

6. Which of the following is (are) illegal?

(a) `LDI R20,500` (b) `LDI R23,50` (c) `LDI R1,00`
(d) `LDI R16,$255` (e) `LDI R42,$25` (f) `LDI R23,0xF5`
(g) `LDI 123,0x50`

7. Which of the following is (are) illegal?

(a) `ADD R20,R11` (b) `ADD R16,R1` (c) `ADD R52,R16`

Solutions

1. 8
2. 8
3. 8
4. 0xFF
5. $28 – in R20
6. (a), (c), (d), (e), (g)
7. (c)

13. True or false. The I/O registers are part of the data memory space.
14. True or false. The general-purpose registers are not part of the data memory space.
15. True or false. The data memory is the same size in all members of AVR.
16. If we add the I/O registers, internal RAM, and general purpose register sizes together we should get the total space for the _____.

Solutions

13. True
14. False
15. False
16. 32 bytes GPR + 64 bytes I/O locations + 2048 (2k RAM) = 2144 bytes

18. What is the difference between the EEPROM and data RAM space in the AVR?
19. Can we have an AVR chip with no EEPROM?
20. Can we have an AVR chip with no data memory?
21. What is the address range for the internal RAM?
22. What is the maximum number of bytes that the AVR can have for the data memory?

Solutions

18. EEPROM does not lose its data when power is off, whereas SRAM does. So, the EEPROM is used for storing data that should rarely be changed and should not be lost when the power is off (e.g., options and settings); whereas the SRAM is used for storing data and parameters that are changed frequently.
19. Yes
20. No, each microcontroller should have general purpose registers and I/O registers.
21. For the ATmega32 ($0000-$085F for 0-2143 from problem 16)

22. 65,536 bytes or 2048 for ATmega32

29. Show a simple code to (a) load the value $15 into location $67, and (b) add it to R19 five times and place the result in R19 as the values are added. R19 should be zero before the addition starts.

31. Write a simple code to complement the contents of location $68 and place the result in R27.

Solutions

29.
```
LDI R16, $15
STS $67, R16
LDI R19, 0
LDS R20, $67
ADD R19, R20
ADD R19, R20
ADD R19, R20
ADD R19, R20
ADD R19, R20
```

31.
```
LDS R27, $68
COM R27
```

38. What is the status of the C and Z flags after the following code?
```
LDI R20,0xFF
LDI R21,1
ADD R20,R21
```

39. Find the C flag value after each of the following codes:

(a)
```
LDI R20,0x54
LDI R25,0xC4
ADD R20,R25
```

(b)
```
LDI R23,0
LDI R16,0xFF
ADD R23,R16
```

(c)
```
LDI R30,0xFF
LDI R18,0x05
ADD R30,R18
```

Solutions

38. C = 1 because there is a carry beyond the D7 bit.
Z = 1 because the R20 (the result) has value 0 in it after the addition.

39.
(a)
$54 0101 0100
+ $C4 1100 0100
$118 10001 1000 R20 = $18
C = 1 because there is a carry beyond the D7 bit.

(b)
$00 0000 0000
+ $FF 1111 1111
$FF 1111 1111 R23 = $FF
C = 0 because there is no carry beyond the D7 bit.
(c)
$FF 1111 1111
+ $05 0000 0101
$FF 10000 0100 R30 = $04
C = 1 because there is a carry beyond the D7 bit.

42. State the value (in hex) for each of the following data:

```
.EQU DAT_1 = 22
.EQU DAT_2 = $56
.EQU DAT_3 = 0b10011001
.EQU DAT_4 = 32
.EQU DAT_5 = 0xF6
.EQU DAT_6 = 0B11111011
```

Solutions

42.

```
.EQU DAT_1 = $16
.EQU DAT_2 = $56
.EQU DAT_3 = $99
.EQU DAT_4 = $20
.EQU DAT_5 = $F6
.EQU DAT_6 = $FB
```

44. Assembly language is a _____ (low, high)-level language while C is a _____ (low, high)-level language.
45. Of C and Assembly language, which is more efficient in terms of code generation (i.e., the amount of ROM space it uses)?
46. Which program produces the obj file?
47. True or false. The source file has the extension "asm".

Solutions

44. Low, High
45. Assembly

46. Assembler
47. True

51. Why are the directives also called pseudocode?
52. True or false. The .ORG directive appears in the ".lst" file.
53. The file with the _____ extension is downloaded into AVR Flash ROM.

Solutions

51. Because they do not produce machine instructions. They just give directions to the assembler.
52. True
53. hex

57. Find the number of bytes each of the following instructions takes:

(a) `LDI R19,0x5`	(b) `LDI R30,$9F`	(c) `ADD R20,R21`
(d) `ADD R22,R20`	(e) `LDI R18,0x41`	(f) `LDI R28,20`
(g) `ADD R1,R3`	(h) `JMP`	

63. Find the on-chip program ROM size in K for the AVR chip with the following address ranges:

(a) $0000–$1FFF	(b) $0000–$3FFF
(c) $0000–$7FFF	(d) $0000–$FFFF
(e) $0000–$1FFFF	(f) $00000–$3FFFF
(g) $00000–$FFF	(h) $00000–$1FF

Solutions

57. a) 2 bytes b) 2 bytes c) 2 bytes d) 2 bytes
e) 2 bytes f) 2 bytes g) 2 bytes h) 4 bytes

63.

(a) $1FFF + 1 = 8,192 words = 16,384 bytes = 16 KB
(b) $3FFF + 1 = 16,384 words = 32 KB
(c) $7FFF + 1 = 32,768 words = 64 KB
(d) $FFFF + 1 = 65,536 words = 128 KB
(e) $1FFFF + 1 = 131,072 words = 256 KB
(f) $3FFFF + 1 = 262,144 words = 512 KB
(g) $FFF + 1 = 4096 words = 8 KB
(h) $1FF + 1 = 512 words = 1 KB

The answers are given in terms of bit words. If considered as 16 bit words, then there is a scale factor of 2.

Additional Problem:

Write a sequence of ATmega32 assembly instructions to perform the following:

a) add 32 to 86
b) store the sum in data memory location 0x120
c) initialize R20 to 0, subtract 5 from R20 five times
d) take the 1s complement of the difference from part c
e) output the 1s complement to PORTB

Solution:

a) add 32 to 86
```
LDI R16, 32
LDI R17, 86
ADD R16, R17; MANY POSSIBLE SOLUTIONS
```
b) store the sum in data memory location 0x120
```
STS 0x120, R16
```
c) initialize R20 to 0, subtract 5 from R20 five times
```
LDI R20, 0

LDI R21, 5; 5 TO SUBTRACT FROM R20

LDI R22, 5; ITERATIONS FOR SUBTRACTION

LOOP: SUB R20,R21; SUBTRACT 5 (SOURCE) FROM R20 (DESTINATION)

        DEC R22; DECREMENT COUNTER

        BRNE LOOP; REPEATED 5 TIMES?
```
d) take the 1s complement of the difference from part c
```
COM R20; TAKE 1S COMP OF DIFFERENCE STORED IN R20
```
e) output the 1s complement to PORTB
```
OUT PORTB, R20; OUTPUT 1S COMP OF DIFFERENCE FROM PART C TO PORTB
```